

Yissum Research Intelligence Report — Exact & Social Sciences — Week 30, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 30, 2026 · Generated July 22, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Hebrew University researchers are leveraging diverse scientific insights to develop cutting-edge solutions for critical industry needs. One project focuses on biomimetic AI, drawing from insect flight dynamics to enhance autonomous micro-aerial vehicle performance in low-visibility conditions. Another groundbreaking study utilizes extensive clinical and environmental data to create predictive tools for personalized prenatal health monitoring. These innovations aim to bring robust control to autonomous systems and foster proactive, data-informed healthcare interventions.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- Tsevi Beatus — Software/AI, Other
- Raanan Raz — Software/AI, Clinical, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Flight in the dark: different responses to darkness in flying insects.

13/50

"Biomimetic AI for MAVs in darkness: Insect-inspired algorithms boost drone stability and control in low-visibility environments."

Main researcher: Tsevi Beatus

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Published · 2026-05

ABOUT THIS RESEARCH

Researchers studied how different flying insects, such as fruit flies, mosquitoes, and honey bees, adapt their flight maneuvers when transitioning from low light to sudden darkness. The study identifies specific kinematic responses, including pitch changes and wingbeat frequency adjustments, that allow certain insects to maintain control without visual cues.

COMMERCIAL ANGLE

The identified flight-stabilization algorithms and sensory-motor delay patterns can be used to develop biomimetic control software for autonomous micro-aerial vehicles (MAVs) operating in low-visibility environments.

COMMERCIALISATION METRICS

Novelty

4/10

The research provides interesting comparative ethology and fine-grained kinematic data, but it is incremental rather than groundbreaking. It describes known biological phenomena in new contexts rather than introducing a novel mechanism, platform, or disruptive technological principle.

Commercial Potential

3/10

The paper is fundamental behavioral biology research that identifies natural flight responses rather than proposing a specific engineering application or biomimetic algorithm. While the data could theoretically inform autonomous drone stabilization, there is no direct translation to a product, service, or licensable technology provided.

Market Size

2/10

The paper is a fundamental biological study of insect behavior with no direct application to a commercial product or industrial market. While it provides basic science, it lacks a defined pathway to a large-scale addressable market or value-driven demand.

Tech Readiness

2/10

This is fundamental biological research focused on understanding insect sensory-motor mechanisms rather than an applied technology or engineering prototype. While the findings could theoretically inform future bio-inspired robotics, the work itself is at the most basic stage of scientific discovery.

IP Strength

2/10

This is a fundamental biological study characterizing natural behavioral responses rather than an invention of a novel device, algorithm, or method. The findings represent scientific knowledge that is difficult to protect via patents and lacks industrial application or a proprietary technological framework.

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2. Associations between prenatal exposure to ambient temperature and birthweight small-for-gestational-age: an integrative model.

12/50

"Climate-aware prenatal monitoring: AI-powered tools predict birthweight risks by integrating environmental temperature data."

Main researcher: Raanan Raz

School of Public Health, Gray Faculty of Medical & Health Sciences, Tel Aviv University, Tel Aviv, Israel.; Braun School of Public Health and Community Medicine, The Hebrew University at Jerusalem - Hadassah, Jerusalem, Israel.

Published in International journal of environmental health research · 2026-04

ABOUT THIS RESEARCH

Researchers analyzed a large dataset of 130,000 births to determine how maternal exposure to ambient temperature during specific pregnancy windows affects birth weight and Small for Gestational Age (SGA) outcomes. The study provides a more nuanced understanding of how climate and temperature fluctuations serve as environmental risk factors for neonatal health.

COMMERCIAL ANGLE

The research supports the development of predictive environmental health software and precision prenatal monitoring tools that integrate local climate data to assess maternal and fetal risk.

COMMERCIALISATION METRICS

Novelty

3/10

The paper performs incremental refinement of existing epidemiological knowledge by including preterm births and specific exposure windows rather than introducing a new biological mechanism or a novel technological platform. It represents an observational refinement of prior art rather than a groundbreaking scientific shift.

Commercial Potential

3/10

The study is an epidemiological observation providing population-level insights rather than a novel therapeutic, diagnostic, or device technology. While it informs public health policy, it lacks a direct path to a proprietary product, service, or licensable platform.

Market Size

3/10

The paper is an observational epidemiological study rather than a commercial product or technology, limiting its direct market size to public health policy and niche academic research. While the underlying problem of neonatal health is significant, the research does not describe a scalable, addressable commercial market or a proprietary intervention.

Tech Readiness

2/10

This is a retrospective observational epidemiological study providing population-level statistical associations rather than a technical intervention or product. It establishes scientific knowledge but does not present a developed technology, tool, or clinical device ready for application.

IP Strength

1/10

The paper presents an epidemiological observational study of environmental correlations, which is inherently non-patentable and lacks any proprietary technology, molecule, or device to defend. The findings constitute scientific knowledge rather than a protectable intellectual property asset.

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